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SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	07 July 2026
Team ID	SWTID-2026-8075
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project, Voice Based Concept Understanding Analyser, is an AI-powered application designed to evaluate a student's conceptual understanding through voice explanations. The system converts speech into text using OpenAI Whisper, compares the explanation with a reference answer using semantic similarity analysis, and evaluates fluency based on speaking speed, pauses, voice energy, and filler words. It provides instant AI-generated feedback along with a downloadable PDF report, helping students improve their understanding, communication skills, and overall learning experience.

Project Overview	
Objective	Develop an AI-powered Voice Based Concept Understanding Analyser that evaluates a student's conceptual understanding through voice explanations. The system automatically converts speech into text, compares it with a reference answer, analyzes speech quality, and generates detailed performance feedback.
Scope	The project enables students to upload voice recordings of concept explanations. It performs speech-to-text conversion, semantic similarity analysis, fluency analysis, pause detection, voice energy analysis, filler word detection, AI-based feedback generation, and PDF report generation for self-learning and educational assessment.
Problem Statement	

Description	Traditional methods of evaluating conceptual understanding are time-consuming, subjective, and require teacher involvement. Students often do not receive instant or detailed feedback about their explanations, making it difficult to identify weaknesses and improve learning.
Impact	The proposed solution provides immediate and objective feedback on conceptual understanding, improves learning outcomes, reduces teachers' evaluation workload, and helps students practice more effectively for exams, presentations, and interviews.
Proposed Solution	
Approach	The system uses OpenAI Whisper for speech-to-text conversion and compares the transcribed explanation with a predefined reference answer using semantic similarity analysis. It further evaluates fluency, speaking speed, pauses, voice energy, and filler words before generating AI-based feedback and a downloadable PDF report.
Key Features	- Speech-to-Text using Whisper AI - Semantic Similarity Analysis - Concept Understanding Score - Fluency Analysis (Words Per Minute) - Pause Detection - RMS Voice Energy Analysis - Filler Word Detection - AI Performance Summary - Downloadable PDF Report

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel i5

Memory	RAM	Minimum 8 GB
Storage	Disk space	2–5 GB Free Storage
Software		
Frameworks	Streamlit	Streamlit
Libraries	Additional Libraries	OpenAI Whisper, Librosa, Matplotlib, NumPy, Sentence Transformers, ReportLab
Development Environment	IDE	Visual Studio Code, PyCharm
Data		
Data	Source, Size, Format	User Voice Recordings, Reference Answers, WAV/MP3/M4A/MP4/OGG/OPUS/FLAC (Max 200 MB), Text Data